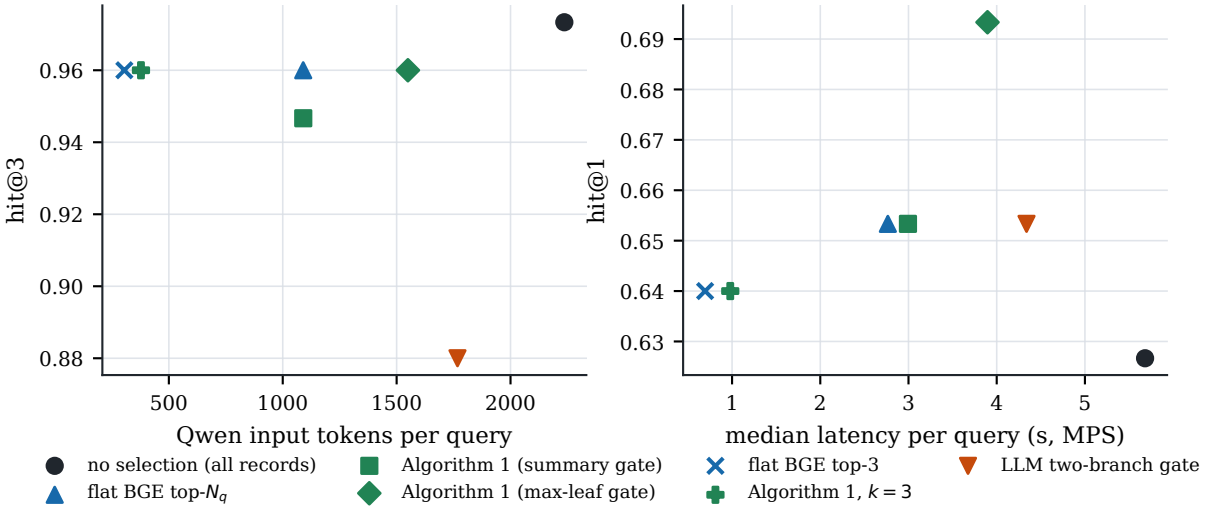
(a) Recall versus context
hit@3
0.96
0.94
0.92
0.90
0.88
500
1000
1500
2000
Qwen input tokens per query
(b) Top-1 accuracy versus latency
hit@1
0.69
0.68
0.67
0.66
0.65
0.64
0.63
1
2
3
4
5
median latency per query (s, MPS)
no selection (all records)
flat BGE top-N_q
Algorithm 1 (summary gate)
Algorithm 1 (max-leaf gate)
flat BGE top-3
Algorithm 1, $k = 3$
LLM two-branch gate